

A feasibility-preserving quantum-classical optimizer for Distributed Order Management

Executive summary

Distributed Order Management (DOM) decides which distribution center (DC) and ship date should serve each customer order. When the default DC cannot fully serve an order, diversion may improve fulfillment and avoid penalties, but it can increase shipping cost or consume inventory and operational headroom needed elsewhere. These interactions make DOM a coupled combinatorial optimization problem rather than a set of independent order rankings.

This submission implements the full workflow on the readable Nestlé proof-of-concept (POC) data. A strict gate opens all ten supplied artifacts and stops on the first unreadable or structurally invalid file. The source adapter derives cases, order economics, thresholded penalties, protected inventory, shipping candidates, load groups, and dock resources. The supplied outputs are used only for reconciliation, never as optimization labels. Restricted rows, identifiers, DC details, and exact commercial totals are excluded from the repository.

The common deterministic model is a mixed-integer linear program (MILP). It selects one eligible DC/planned-goods-issue (PGI) option per order—or an explicit unassigned outcome—and chooses integer SKU quantities. It enforces demand balance, one-DC fulfillment, assignment-group cohesion, projected available-to-promise (ATP) inventory, dock and optional scenario resources, exact pallet/loose-case accounting, and the POC diversion-uplift rule. Its objective is fulfilled value minus thresholded unmet penalty minus total shipping cost.

Two transparent baselines provide fair controls. The default baseline keeps loads at their original option. The sequential greedy baseline previews alternatives against current residual resources, commits the best incremental choice, and updates the remaining state. SciPy/HiGHS solves the exact model on tractable subsets and reports a mathematical bound and optimality gap.

The hybrid method is a feasibility-preserving large-neighborhood search (LNS). It freezes most orders, selects a bounded set of resource-conflicting loads, and builds a one-hot quadratic unconstrained binary optimization (QUBO) model over assignment plans. Exact enumeration, random sampling, simulated annealing, or an approved D-Wave backend can propose combinations. Deterministic repair restores one common option per load. Exact local MILP recourse rebuilds SKU quantities using residual resources, and an independent global validator accepts only a strict feasible improvement. The QUBO is therefore a proposal mechanism, not the feasibility authority.

The notebook implements the required comparison and all proposed experiments: size scaling, penalty sensitivity, candidate sensitivity, inventory shocks, seed and local coefficient-noise sensitivity, Pareto-pruning ablation, and random-versus-conflict batching. It adds random-versus-annealing sampling and a synthetic coordination control. The executed smoke profile produced 37 aggregate runs, all feasible, with no hybrid regression from its configured incumbent. On the smallest real subset the exact MILP proved a zero gap, and greedy and hybrid matched it. On a larger subset the safe smoke hybrid improved its default start but did not reach greedy/exact. This is honest evidence for architectural safety, not quantum advantage. No QPU run or quantum-advantage claim is made.

1. Business problem and decision value

An order normally ships from its default DC. If the default cannot fully serve its SKUs, a planner may move the load to an alternate DC. A better fill can preserve sales and customer service, but the decision has network consequences:

- an alternate lane may cost more;
- stock used now may be protected for future demand;
- an alternate shipment may require scarce dock headroom;
- a source load may contain multiple orders that must stay together;
- a partial fill can trigger an order-level shortage penalty; and
- a candidate that looks good alone may conflict with other recommended candidates.

If each of O decision units exposes K options, the assignment space is approximately K^O . Each assignment then induces integer fulfillment quantities for its SKU lines. Inventory couples orders sharing a DC and SKU, while dock or pick resources couple otherwise unrelated SKUs. Load cohesion further links orders that would be independent in a simple assignment problem.

The planner needs more than a selected DC. A usable recommendation must explain its fill change, penalty avoided, shipping change, service date, binding resource, and net objective effect. It must also preserve a safe fallback and state any assumption that still needs operational confirmation.

2. Data understanding and strict ingestion

2.1 Bundle inventory

All supplied files are readable. The challenge bundle contains:

Source	Size	Use
Order-SKU input	25,193 rows	Demand, default assignment, dates, values, penalty parameters, units, load, priority
Capacity-planning input	377,504 rows	Daily DC-SKU inventory and forecast availability
Shipping input	12,922 rows	Plant-to-destination lane cost and distance
Dock input	480 rows	Date-specific remaining dock availability
Throughput input	530 rows	Observed case/pallet/order utilization
Equations DOCX	110 equation objects	Source rule definitions
Example workbook	561 rows	Worked POC calculation
Order output	1,109 rows	Aggregate and order-level reconciliation only
SKU output	25,193 rows	Line-level reconciliation only
Challenge PDF	5 pages	Tasks, criteria, privacy, and submission contract

The format gate parses CSVs, opens the workbook, validates the Word OOXML package, and extracts every PDF page. It raises before model construction if any artifact fails. This implements the explicit requirement to stop rather than guess from an unreadable source.

2.2 Quantity conversion

The order input is one row per order-SKU in source planning units. Integer case demand is

$$Q_{os} = \text{round}(U_{os}/C_{os})$$

when positive planning units per case C_{os} exist. For pallet planning-unit rows it is

$$Q_{os} = \text{round}(U_{os}P_s),$$

where P_s is cases per pallet. The adapter rejects a result that is not integral within tolerance. This rule exactly reconciles to the supplied output case demand. Unit value, weight, and volume are source totals divided by cases.

2.3 Thresholded penalty

The equations and output show an order-level penalty, not a penalty charged on every unmet case regardless of fill. Let θ_o be fill threshold. Penalty is inactive once integer filled cases reach $\lceil \theta_o Q_o \rceil$. Below the threshold:

$$R_o = F_o + \sum_s \pi_{os} u_{os} + K_o \sum_s \mathbf{1}[u_{os} > 0].$$

The active amount is raised to an optional minimum and clipped to an optional positive maximum. π_{os} is case value times the source shortage factor. The evaluator reproduces `PenaltyIfNotDiverted` with maximum absolute error below 4.5×10^{-7} , providing a strong equation-level audit.

2.4 Focus population and load cohesion

Rows with unavailable default inventory identify focus orders. If one order on a source load is a focus order, the entire load enters the population so the solver can preserve routing cohesion. Missing load identifiers become one-order singleton groups. The source contains 146 multi-order loads, and the supplied recommendation never splits one load across DC/date. With the default adapter settings, focus expansion produces 750 orders, 20,869 order-SKU lines, and 372 assignment groups.

2.5 Protected ATP

The inventory table satisfies

$$\text{Available inventory} = \text{Opening stock} - \text{Reserved quantity}.$$

Negative values are clipped to zero. For candidate PGI t , the solver uses the minimum availability over t through $t + 5$ calendar days. This protects already committed future consumption. An alternative DC must contain every SKU in the load's inventory/forecast table. Absence at the default DC means zero preview fill, not a missing default option.

2.6 Lanes, dates, and resources

Shipping lead is $\text{ceil}(\text{distance} / 500)$ calendar days. Alternative PGI is requested delivery date minus lead time, moved backward over weekends and configured holidays. Lane cost is treated as total option cost. Shipping and dock use are charged once via a deterministic group leader.

Dock_Remaining is clipped at zero and is the only real operational capacity limit whose remaining amount is explicit. Default loads consume zero incremental dock; alternate loads consume one. The throughput file reports observed utilization but no documented maximum. The adapter therefore does not silently convert it into a hard constraint. A user may request an explicitly labeled headroom scenario.

2.7 Candidates and pruning

Before pruning, the default focus universe exposes 2,182 order candidate rows and 47,075 protected-ATP rows. An option is retained only when every member of a load can use it. Pareto pruning removes an alternate only if another option has no worse estimated fill and fulfilled value, no greater shipping cost and lead time, and is strictly better in at least one dimension. Default options are never removed. The result has 1,307 candidate rows.

3. Deterministic mathematical model

Let x_c select candidate c , z_o represent no assignment, f_{osc} be fulfilled cases, and u_{os} be unmet cases. The objective is

$$\max J = \sum_{o,s,c} v_{os} f_{osc} - \sum_o P_o - \sum_c c_c x_c.$$

Each order has one modeled outcome:

$$\sum_{c \in \mathcal{C}_o} x_c + z_o = 1.$$

Demand and assignment linking are

$$\sum_{c \in \mathcal{C}_o} f_{osc} + u_{os} = Q_{os}, \quad 0 \leq f_{osc} \leq Q_{os} x_c.$$

For assignment group g , every member selects the same shared option as its leader, or all are unassigned. This is encoded by equality constraints on corresponding candidate binaries and unassigned binaries.

Protected ATP at every checkpoint is

$$\sum_{o,c:d(c)=d,t(c) \leq t} f_{osc} \leq I_{dst}.$$

For resource r , fixed candidate use β_c^r , variable use α_{osc}^r , and limit R_{dt}^r :

$$\sum_{c:d(c)=d,t(c)=t} \beta_c^r x_c + \sum_{o,s,c:d(c)=d,t(c)=t} \alpha_{osc}^r f_{osc} \leq R_{dt}^r.$$

When pallet/pick scenarios are enabled, exact case decomposition is

$$f_{osc} = P_s p_{osc} + k_{osc}, \quad 0 \leq k_{osc} \leq (P_s - 1) x_c.$$

A non-default candidate must fill at least

$$L_o^{\text{div}} = \min\{Q_o, F_o^{\text{def}} + \max(\lceil 0.05 Q_o \rceil, 100)\}.$$

Threshold activation, unmet-active products, cut-SKU indicators, and floor/cap selection use bounded big-M linearizations. Bounds are order-specific. The exact implementation and independent objective evaluator agree within numerical tolerance.

SciPy submits this MILP to HiGHS. A zero optimality gap proves the returned solution is best for that instance. A time-limited solve may return a strong feasible incumbent and a bound without proof.

4. Classical baselines

4.1 Default assignment

The default baseline limits each load to its original option. It allocates quantities against shared protected ATP and capacity in deterministic order and applies the same penalty equation and validator as every other method. This answers the operational question: what is achievable without diversion under the modeled resources?

4.2 Sequential greedy reassignment

The greedy baseline previews each currently eligible group option, scores its incremental business objective, commits the best feasible outcome, and immediately updates residual resources. Ties prefer larger objective, larger fill, lower shipping cost, default source, then deterministic candidate key.

Greedy is explainable and fast. Its limitation is path dependence: after it commits a load, it cannot revisit that decision when another load later reveals a better joint combination. This makes it an important control for large-neighborhood search.

4.3 Exact MILP reference

The MILP is the quality reference on tractable subsets and the fulfillment recourse engine inside the hybrid. Its recorded diagnostics include variable and constraint counts, node count, incumbent, upper bound, optimality gap, and wall time. Exact and heuristic rows are evaluated by the same external code.

5. Hybrid quantum-classical design

5.1 Why not a monolithic QUBO

A direct QUBO would need binary encodings for SKU quantities, inventory and capacity slack, thresholded penalty activation, floor/cap logic, and load cohesion. It would have far more logical variables than the assignment problem, broad coefficient ranges, many penalty-calibration risks, and embedding overhead on sparse hardware. Current hardware is better suited to a bounded binary selection problem than the full operational model.

5.2 Bounded neighborhood

The hybrid starts from a feasible default or greedy incumbent. It builds a conflict view from shared DC-SKU inventory and DC-date resources. The conflict strategy selects complete groups with unmet demand, high conflict degree, and useful alternatives. The random-batch ablation selects complete groups by seed. Both active orders and logical variables are capped.

Frozen-order consumption is subtracted from every inventory checkpoint and capacity bucket. This residualization prevents local recourse from reusing resources already committed outside the neighborhood.

5.3 Assignment-plan QUBO

For active order o and preview plan k , binary y_{ok} selects the plan. The QUBO minimizes

$$E(y) = - \sum_{o,k} \tilde{w}_{ok} y_{ok} + P_{\text{one}} \sum_o \left(1 - \sum_k y_{ok} \right)^2 + \sum_{(ok, o'k')} P_{ok, o'k'} y_{ok} y_{o'k'}.$$

The first term rewards isolated business value, the second encourages one outcome per order, and the third penalizes candidate pairs contributing to shared-resource contention. A higher-order pressure estimate adds signal when several individually compatible plans overload a resource collectively. This remains a surrogate; exact recourse resolves all real constraints.

The incumbent supplies a one-hot warm start. Available local samplers are exact enumeration for tiny QUBOs, seeded random sampling, and simulated annealing. Optional D-Wave QPU and managed-hybrid adapters require credentials and explicit data approval.

5.4 Repair, recourse, and safety

Raw samples can violate one-hot structure. Repair selects one plan per order and projects all members of a load onto one common option. The best unique repaired assignments are fixed in the exact local MILP. The MILP reoptimizes SKU quantities under residual resources; infeasible proposals are discarded.

The local solution is merged with frozen orders. An independent validator then checks assignment uniqueness, load cohesion, demand identity, candidate eligibility, date/DC consistency, protected ATP, enabled capacities, and diversion uplift. The objective is recomputed. A proposal replaces incumbent S^k only if

$$\widehat{S} \text{ is feasible} \quad \wedge \quad J(\widehat{S}) > J(S^k).$$

Therefore $J(S^{k+1}) \geq J(S^k)$. A bad sampler can waste a call, but it cannot degrade the returned recommendation.

6. Evaluation design

Every experiment is defined by bundle hash, schema and assumption versions, candidate set, common objective, method configuration, seed, and commit. Whole-load expansion means an 8-order request can contain more than eight actual orders; both counts are reported.

The full profile contains:

1. a default/greedy/exact/hybrid solver comparison;
2. real-data size scaling at requested sizes 8, 20, and 50;
3. penalty scales 0.5, 1.0, and 2.0;
4. candidate limits 1, 2, 4, and 6;
5. inventory reductions 0%, 10%, and 25%;
6. seeds 3, 11, 29, and 47 with QUBO coefficient noise 0%, 1%, and 5%;
7. Pareto pruning off/on;
8. random/conflict batch selection;
9. random/simulated-annealing sampler comparison; and
10. an eight-order synthetic coordination control.

The smoke profile reduces levels, sampler reads, sweeps, and hybrid iterations for development. All persisted rows contain aggregate metrics only.

Noise perturbation is reproducible symmetric Gaussian variation of local QUBO coefficients. Samples are ranked on the original model and passed through exact recourse. This tests coefficient sensitivity; it does not model analog control error, embedding chains, readout, thermal effects, or gate noise.

7. Results and interpretation

7.1 Supplied-output audit

The reference output contains 1,109 orders, 25,193 order-SKU rows, 614 named loads, 631 assignment groups after singleton handling, and three diversions. Selected fill is 2,413,937 of 2,554,440 cases, or 94.4997%. Order/SKU coverage is complete and the source penalty equation reconciles. These facts validate transformation logic; they do not establish that the historical recommendation is optimal.

7.2 Executed smoke profile

The smoke notebook produced 37 aggregate experiment rows. Every row passed independent validation and no hybrid row returned an objective below its starting default incumbent.

For the common four-decision-unit real subset, normalize the default objective to 100. Greedy, exact MILP, and hybrid each reached 136.5. The exact MILP gap was zero. The hybrid used a maximum of nine logical QUBO variables and accepted one move. This shows that the workflow can recover and certify the best subset solution, but it does not show a sampler advantage because greedy also reached it.

For the requested eight-order scaling point, whole-load expansion produced nine orders in eight groups. Greedy and zero-gap exact MILP agreed. The one-iteration smoke hybrid improved its default incumbent but remained below them. The result is valuable because it demonstrates the safety invariant under an insufficient search budget rather than hiding a negative outcome.

With Pareto pruning, the base subset candidate rows fell from 12 to 6 while the validated objective remained unchanged. Random and conflict batches tied on the tiny base, as did random and simulated-annealing sampling; that instance is too small to differentiate them. The four seed/noise smoke rows all remained feasible, while one 2% perturbation produced a smaller improvement, confirming ranking sensitivity.

In the independent synthetic coordination control, greedy scored -3895.8 synthetic units, exact MILP proved -3623.4 , and hybrid reached -3872.8 . Hybrid improved greedy by 23 units but remained 249.4 below exact. This supports the value of revisiting coupled assignments while making clear that the current configuration is not dominant.

7.3 Known-optimum unit test

The two-order synthetic test has proven objective 126, with o_1 assigned to d_2 and o_2 assigned to d_1 . Exact MILP and exact-QUBO hybrid reproduce it from a default objective of -50 . The example is synthetic and serves as a correctness gate.

8. Scaling, robustness, and quantum scope

For active batch B with candidate plans $\mathcal{K}_o$, logical QUBO width is

$$n_{\text{QUBO}} = \sum_{o \in B} (|\mathcal{K}_o| + 1).$$

Pair construction is worst-case $O(n_{\text{QUBO}}^2)$. The configured variable cap bounds the sampler problem even when global order volume grows. Outer candidate generation, conflict selection, and repeated recourse still grow and are the practical classical costs.

Global MILP size grows with candidate assignments plus order-line-candidate quantity arcs; thresholded penalties and exact pick accounting add auxiliaries. Global exact optimization is therefore a subset reference or time-limited method. Local recourse fixes assignment binaries and is materially smaller.

Production improvements include cached inventory profiles and plan signatures, Pareto and candidate-column reduction, dual-price or reduced-cost neighborhood selection, parallel neighborhoods against versioned incumbents, adaptive batch size, and a classical-only fallback for every remote call.

Physical qubit count is not logical QUBO capacity. Minor embedding, chain length, connectivity, coefficient range, anneal schedule, queue time, and post-processing affect usable scale. A fair QPU study must compare identical QUBOs, warm starts, repair, recourse, and validation under both equal time and equal quality. It must report preprocessing, embedding, queue, QPU access, sampling, repair, and recourse time separately across multiple seeds. The present work does not make that claim.

9. Planner copilot and operational use

The Streamlit planner copilot is justified because the evidence spans methods, sizes, economics, scarcity, sampler settings, and ablations. It presents aggregate charts and answers bounded questions such as “Did Pareto pruning help?” and “Is this quantum advantage?” It calls no external language model and rejects columns named like order, SKU, DC, customer, candidate, or ZIP identifiers.

The copilot is not a routing agent. A private planner export must still show each load’s default and recommended option, fill change, penalty avoided, shipping change, net change, service date, and binding reason. A planner verifies extraction time, holiday calendar, load rule, dock semantics, commercial scale, and validator status. Manual overrides are recorded with their missing business rule.

10. Limitations and next steps

The complete holiday calendar is not supplied. Throughput rows are observed utilization, not a documented maximum. Customer-specific all-or-nothing service is not modeled because the challenge allows partial fulfillment and no source class identifies exceptions. Commercial scale and an external QPU data boundary require organizer confirmation.

The QUBO uses previewed plan value and quadratic contention; exact aggregate feasibility remains in recourse. Candidate preview may not capture every joint effect, and coefficient scaling may need hardware tuning. The current conflict selector does not use MILP dual prices. The full-profile notebook is implemented but should be rerun in the approved environment for final timing and uncertainty plots.

Recommended next steps are:

1. confirm the plant working-day calendar and throughput-limit equation;
2. run the full profile and retain aggregate evidence with commit and bundle hash;
3. increase hybrid iterations or use greedy initialization on larger subsets;
4. add dual-informed batch selection and adaptive recourse budgets;
5. have planners review private row-level explanations; and
6. run an approved QPU comparison only after tuned local controls are fixed.

11. Conclusion

The submission turns the readable POC into a reproducible and independently validated DOM optimization workflow. Exact MILP retains responsibility for quantities and hard operational constraints. The QUBO is confined to a bounded, coordinated assignment search where quantum sampling could be compared fairly. This partition is scalable, privacy-aware, and robust to weak samples.

The current evidence supports the architecture and its safety invariant, not a claim that quantum hardware outperforms classical optimization. That distinction is a strength: planners keep a certified classical reference and fallback, while the project exposes a controlled place for future hardware experiments.

References

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